

DMS613 PROJECT

How Close Are the Option Pricing Formulas
of Bachelier and Black-Merton-Scholes?

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INTRODUCTION & MOTIVATION

ISSUE IN BACHELIER

Bachelier's original framework uses arithmetic Brownian motion, which has a theoretical flaw of allowing negative stock prices.

RECTIFICATION BY BMS

The BMS framework rectifies this issue by using a lognormal assumption.

SIMILARITY

Despite this theoretical flaw, the practical discrepancy between the two models is surprisingly small for short-dated options.

CORE PRICING FORMULAS

Both frameworks define the pricing for a European Call option at time $t=0$ with maturity T and strike K .

The Bachelier call price equation

$$C_0^B = (S_0 - K) \Phi \left(\frac{S_0 - K}{\sigma^B \sqrt{T}} \right) + \sigma^B \sqrt{T} \phi \left(\frac{S_0 - K}{\sigma^B \sqrt{T}} \right)$$

where

$\Phi(\cdot)$: The standard normal cumulative distribution function (CDF)

$\phi(\cdot)$: The standard normal probability density function (PDF)

σ^B : The Bachelier volatility parameter,

The BMS call price equation

(for $r=0$)

$$C_0^{BS} = S_0 \Phi(d_1) - K \Phi(d_2)$$

where

$$d_1 = \frac{\ln(S_0/K) + \frac{1}{2}\sigma^2 T}{\sigma \sqrt{T}} \text{ and } d_2 = d_1 - \sigma \sqrt{T}$$

SYSTEM DYNAMICS AND SDES

The fundamental divergence of the models is best illustrated by their underlying Stochastic Differential Equations (SDEs).

BACHELIER

It uses additive dynamics, meaning the price changes by a flat amount similar to simple interest.

$$dS_t^B = \sigma^B dW_t$$

BSM

It uses multiplicative dynamics, meaning the price changes proportionally similar to compound interest

$$dS_t^{BS} = S_t^{BS} \sigma dW_t$$

By enforcing the volatility scaling $\sigma^B = S_0 \sigma$, the two SDEs become virtually identical in the short run because the stock price stays close to its starting point.

THE MARTINGALE PROPERTY

- Bachelier operated on the "fundamental principle" that a speculator's mathematical expectation is zero.
- In modern financial terms, this means the discounted price process must be a martingale under the pricing measure.
- **Bachelier:** $S_T = S_0 + \sigma^B W_T$. Since the expectation of a Brownian motion increment W_T is zero, $E[S_T] = S_0$.
- **BMS (with $r = 0$):** $S_T = S_0 \exp(\sigma W_T - \frac{1}{2}\sigma^2 T)$. Since the exponential term forms an exponential martingale, $E[S_T] = S_0$.

CUBIC SCALING

- For At-The-Money (ATM) options where $S_0=K$, the absolute pricing difference between the models can be bounded mathematically.
- The analytical bound for the absolute difference is:
$$0 \leq C_0^B - C_0^{BS} \leq \frac{S_0}{12\sqrt{2\pi}} \sigma^3 T^{3/2}$$
- The relative error can be bounded by:
$$\frac{C_0^B - C_0^{BS}}{C_0^B} \leq \frac{T}{12} \sigma^2$$
- This demonstrates that the ATM pricing difference vanishes at a rate of $O(T^{3/2})$.

PROOF OF THE CUBIC SCALING RATE

- For ATM options ($S_0 = K$) and $r = 0$, the BMS boundaries simplify to $d_1 = \sigma\sqrt{T}/2$ and $d_2 = -d_1$.
- Evaluating the integral of the normal density, the Bachelier ATM price simplifies to:

$$C^B = \sigma^B \sqrt{T} \phi(0) = \frac{S_0 \sigma \sqrt{T}}{\sqrt{2\pi}}$$

- Using the symmetry of the normal distribution where $\Phi(-x) = 1 - \Phi(x)$, the BMS price becomes:

$$C^{BS} = S_0 [\Phi(d_1) - \Phi(-d_1)] = S_0 \left[2\Phi\left(\frac{1}{2}\sigma\sqrt{T}\right) - 1 \right]$$

- We can expand $\Phi(z)$ around $z = 0$ using the Taylor series $\Phi(z) \approx \frac{1}{2} + \frac{z}{\sqrt{2\pi}} - \frac{z^3}{6\sqrt{2\pi}}$. Substituting $z = \frac{1}{2}\sigma\sqrt{T}$:

$$C^{BS} \approx S_0 \left[2 \left(\frac{1}{2} + \frac{\sigma\sqrt{T}}{2\sqrt{2\pi}} - \frac{\sigma^3 T^{3/2}}{48\sqrt{2\pi}} \right) - 1 \right] = \frac{S_0 \sigma \sqrt{T}}{\sqrt{2\pi}} - \frac{S_0 \sigma^3 T^{3/2}}{24\sqrt{2\pi}}$$

- Subtracting the BMS price from the Bachelier price gives the analytical bound:

$$|C^B - C^{BS}| \approx \frac{S_0 \sigma^3 T^{3/2}}{24\sqrt{2\pi}}$$

This demonstrates that the pricing difference is of order $T^{3/2}$. For small T , raising the maturity to the power of 1.5 causes the error to shrink exponentially faster than the price itself.

INTUITION BEHIND THE RESULT

- Bachelier model: $dS_t^B = \sigma^B dW_t$
- Black-Scholes model: $dS_t^{BS} = S_t^{BS} \sigma dW_t$
- Key observation for small T: $S_t \approx S_0$
- Therefore: $S_t \sigma \approx S_0 \sigma = \sigma^B$
- Hence Multiplicative $\approx$ Additive (locally)

BACHELIER'S EXPANSION TECHNIQUES

- Bachelier had to price strikes away from the money without any modern computational tools.
- He defined the call price as a function of moneyness $m = K - S\{0\}$.
- Using a Taylor series and successive derivatives, he developed the following expansion:
$$C(m) = a - \frac{m}{2} + \frac{m^2}{4\pi a} - \frac{m^4}{96\pi^2 a^3} + \dots$$
- This series is a Taylor expansion in the dimensionless quantity m/a .
- Because the Gaussian density is an entire function, this series successfully converges for all values of m .

BACHELIER'S QUADRATIC APPROXIMATION

- Bachelier truncated his series after the quadratic term, resulting in:

$$\hat{C}(m) = a - \frac{m}{2} + \frac{m^2}{4\pi a}$$

- In the 1900 Paris Bourse, option premiums were typically fixed by convention, while the strike prices fluctuated to clear the market.
- Traders needed to calculate the resulting strike price for a given premium.
- This quadratic polynomial provided a highly accurate, tractable equation that traders could invert using the standard quadratic formula.

NUMERICAL VERIFICATION RESULTS

Table 1: Pricing Discrepancy Across Maturities ($S_0 = 100, K = 100, \sigma = 20\%$)

Days (T)	Bachelier Price	BS Price	Abs Diff	Rel Error (%)
1	0.417632	0.417630	0.000002	0.000457
7	1.104950	1.104915	0.000035	0.003196
14	1.562635	1.562535	0.000100	0.006393
30	2.287464	2.287151	0.000313	0.013699
60	3.234963	3.234076	0.000886	0.027398
90	3.962004	3.960376	0.001628	0.041098
180	5.603119	5.598518	0.004602	0.082199
365	7.978846	7.965567	0.013278	0.166694

- For a 1-day maturity, the absolute difference is 0.000002, representing a relative error of roughly 2 parts per million.
- For a 1-year (365 days) maturity, the relative error remains below 0.2 percent.

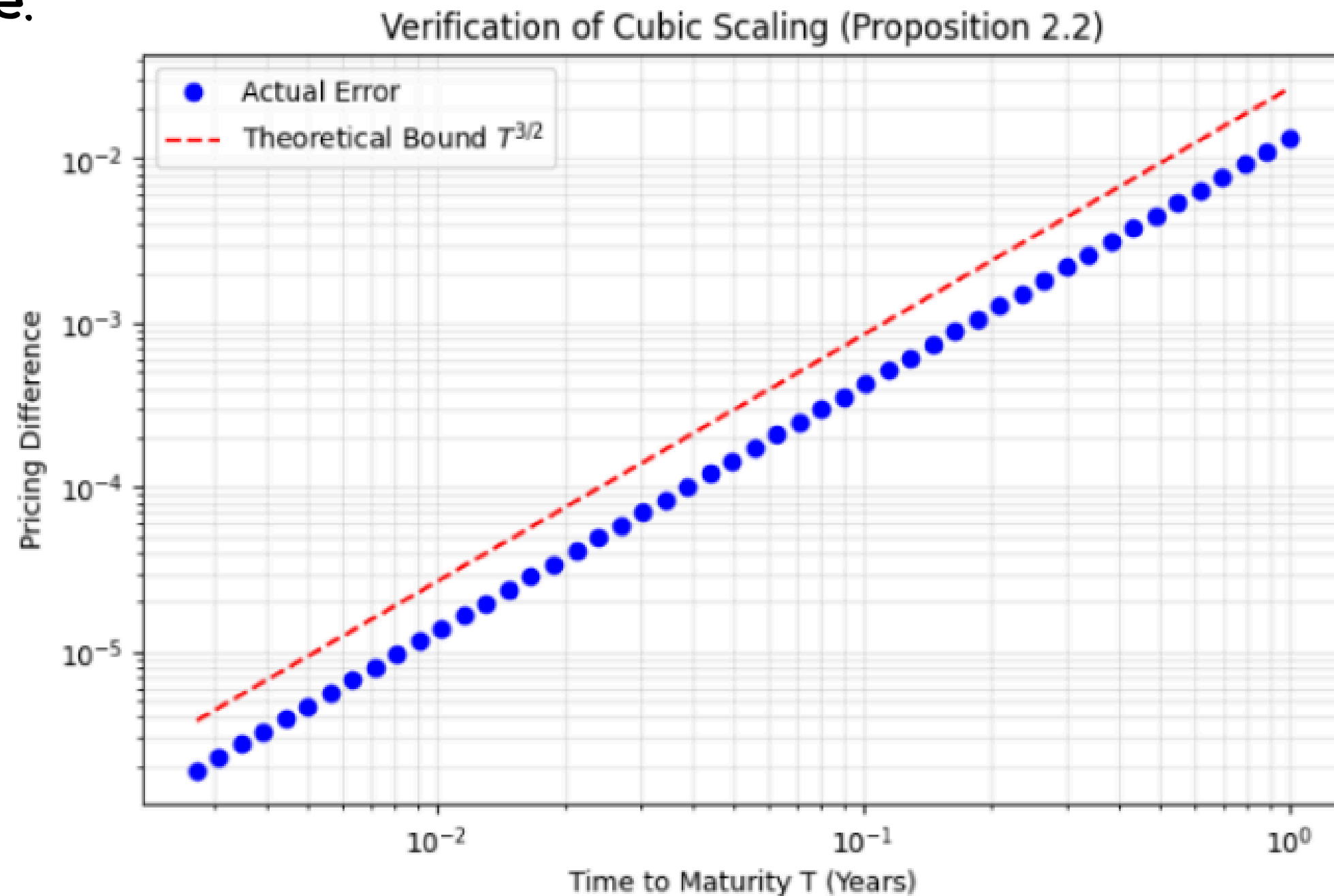
HISTORICAL DATA COMPARISON

Applying the original parameters from the 1900 Paris Bourse ($S = 100$, $\sigma = 2.4\%$, $T = 1/12$ years):

- Bachelier price: 0.2763953196
- Black-Scholes price: 0.2763947668
- Absolute gap: 0.0000005527
- Relative gap: 0.0002%
- It is significantly smaller than the tick size of the exchange, confirming the models were practically identical for historical valuation purposes.

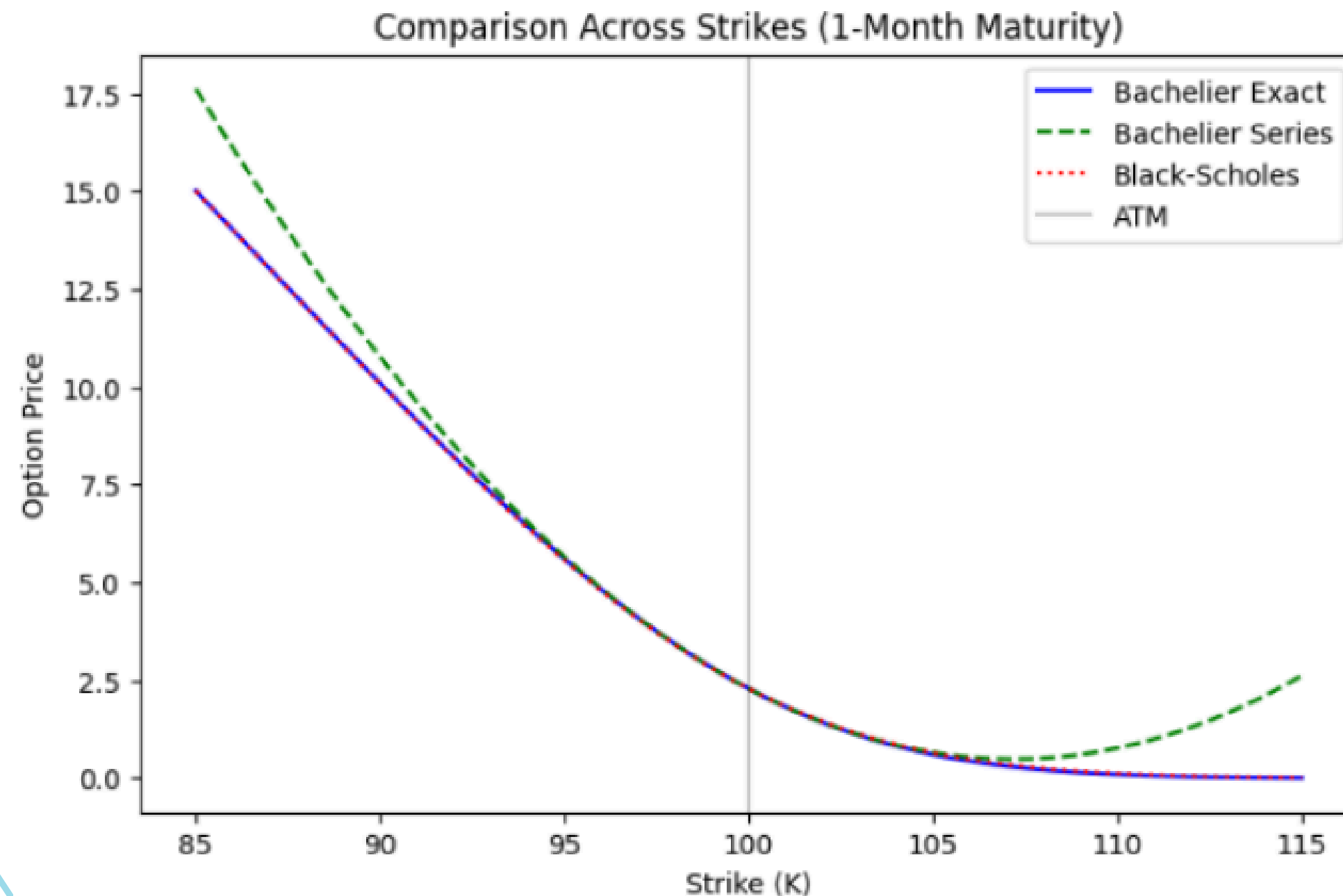
VISUAL VERIFICATION

- This figure depicts the relationship between Time to Maturity (T) and the pricing difference on a log-log scale. The actual error follows the $T^{3/2}$ bound line.



VISUAL VERIFICATION

- **Strike Sensitivity:** This plot compares the exact Bachelier and BS prices against Bachelier's 3-term series approximation. The curves are indistinguishable near the money.



CONCLUSION

- This study confirms the mathematical consistency between the Bachelier and BMS models for short-term options.
- The modern lognormal model is absolutely required for long-dated instruments to prevent negative asset prices.
- However, Bachelier's additive dynamics and quadratic approximations remain incredibly robust and valid for short-term valuations.

“Old model is still surprisingly accurate.”

The background features abstract geometric elements in teal and light blue. In the top-left corner, there are nested rectangular outlines and a diagonal line. The top-right corner contains a grid of small teal circles. The bottom-left corner has a solid teal triangle and another grid of small teal circles. The bottom-right corner shows nested rectangular outlines and a diagonal line.

THANK YOU